

Recoverable Reading Order in Local Research Tools

A. Example · B. Sample · Anonymous synthetic manuscript

Abstract

We test whether document parsers preserve reading order, structure, and citation-ready locations.

1 Introduction

Research readers move repeatedly between a claim and its source. A useful parser therefore preserves both readable prose and the place where the prose appeared.

Two-column papers are a representative stress case because the visual order can differ from the order of drawing commands in the PDF stream.

The benchmark treats a fluent but interleaved paragraph as a structural failure, not as a cosmetic defect.

2 Evaluation protocol

Each candidate returns ordered blocks, headings, page regions, tables, formulas, figures, and an explicit outcome classification.

Human reviewers compare those objects with a small gold set. They may mark a checkpoint as pass, partial, or fail, but cannot average away fabricated content.

A candidate that cannot process a page should report the limitation without inventing text.

¹ Footnote: all names, measurements, and citations in this paper are synthetic.

3 Results

Table 1. Candidate outcomes on representative structures

Candidate	Order	Table	Locator
Text-only	Partial	Fail	Page only
Layout-aware	Pass	Pass	Region
OCR-first	Partial	Partial	Region

Equation 1. Weighted structural score

$$Q = 0.4R + 0.35C + 0.25D$$

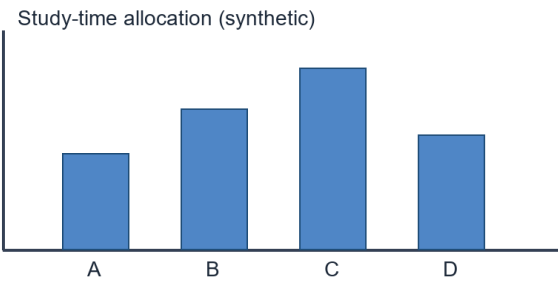


Figure 1. Synthetic quality profile

References

- [1] Example, A. (2024). Reading order as evidence. *Synthetic Review* 4(1), 1-12.
- [2] Sample, B. (2025). Region-level citation locators. *Fictional Computing* 9(3), 22-39.
- [3] Collector Project. (2026). Issue 122 anonymous gold-set protocol.